Prepare DP1 paper (SP-2021)

[SP-2697] Respond to journal review feedback Section 1 and 2

Created: 15/Oct/25 Updated: 12/Feb/26 Resolved: 12/Feb/26

Status:	Done
Project:	System Performance
Components:	None
Fix versions:	None
Parent:	Prepare DP1 paper

Type:	Story	**Priority:**	Undefined
Reporter:	Leanne Guy	**Assignee:**	Leanne Guy
Resolution:	Done	**Votes:**	0
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Remaining Estimate:	Not Specified		
Time Spent:	Not Specified		
Original estimate:	Not Specified		

Reviewers:	James Mullaney
Sprint:	

Description

Respond to review feedback Section 1 and 2

Comments

Comment by Leanne Guy [24/Oct/25]

Section 1:

Line 266 - Currently DP1 is under a two-year proprietary embargo period. How is the DP1 data going to be made accessible to the public? What is the delivery method?

Section 2:

Some more detail about the observing campaign, how locations on sky were selected, and how ComCam observations were deemed "most useful for the community" to be included in DP2 is needed in this section. I provide some specifics questions and comments below.

Add a summary of how the observations were taken, the planned cadence of observations, filters available. Some understanding of the time separation between observations on-sky in the same night, and how often observations were repeated would be useful in addition to the contents of Table 3 and Figures 4 and 5. Is AOS commissioning mostly taking images at the same spot across different airmasses?

Please include the actual dates of the on-sky observing campaigns (start and end of LSSTComCam on-sky observations) early on in this Section rather than the reader having to convert the MJD dates in Figure 5 to get to UT calendar dates.

Why were the specific pointings selected to make DP1? To test the Solar System Processing pipeline (I presume as a low ecliptic field), Seagull and 47 Tucanae to test dense sky backgrounds? Some clarification would be helpful to those who will be using this dataset in the future. How do these fields collectively span the full breadth of the four primary LSST science themes?

Figure 5 - I am finding Figure 5 difficult to parse when comparing to Table 1. From my count it looks like Rubin SV 38 7 has only 10 observations for the number of markers I can count, but Table 1 shows that 159 observations were taken. It also looks like SV 95 -25 gets most of the observations from Figure 5 but from Table 1 it is actually ECDFS that gets the most observations.

Figure 3 - The transmission curves for ComCam are only provided in this paper. The values should be also provided in a supplemental data files or as data behind the figure.

Line 549 - "..accumulating over 200 visits per to evaluate deep coadded images with integrated exposures roughly equivalent to those of the planned LSST Wide Fast Deep 10-year depth" - What are those expected depths? It would be helpful to remind the reader so they can compare to the limiting magnitude of the coadded observations in Table 2.

Table 2 - it would be helpful to add in what the total on-sky exposure time is for the co-added observations per filter per field.

A median 5-sigma per field per filter single PVI (processed visit image) source detection limits/limiting magnitude should be provided in a table.

It would be helpful to provide in some table per field per filter what the saturation apparent magnitude is.

The galactic and ecliptic latitudes for each field should be listed in Table 1 or a separate table.

What were the center pointings for the 4 Rubin_SV_38_7 field pointings? Was the center reported in Table 1 then offset by ~0.65 degrees with small random translational dithers?

It would be useful to discuss the airmass distribution of the visits in the text. Maybe the min/max airmass per filter per pointing could listed in a table or at

Figure 6 is not referenced in the text. This should be presented/discussed in Section 2.4

Section 2.5 - Only median values and best value for seeing were reported in this section. What's the distribution of delivering seeing quality per filter per field? I see some of this information per filter is actually available in Figure 7, but Figure 7 is not referenced in the text. It would be helpful to get a sense of what the required delivering seeing for Rubin Observatory science observations to give a sense of the quality of the data/science return for DP1 compared to the LSST.

I presume fraction of sensors means over CCDs rather than the whole focal plane. Please clarify in the caption for Figure 7.
The cumulative seeing plot per filter is useful, but please also add a histogram of the seeing distribution as a non-cumulative distribution as well.

Comment by Leanne Guy [24/Oct/25]

Additional “minor comments” to address

- Several times in the paper Data Preview (DP2) is referenced as planed to be released in mid-2026. My understanding from reviewing the Rubin Community Workshop presentations (https://rubinobservatory.org/events/rcw-2025) and the Schwamb et al (2025; https://ui.adsabs.harvard.edu/abs/2025RNAAS...9..244S/abstract) is that this is potentially no longer the case given the lessons learned for DP1 processing. The community will try to read tea leaves from the paper text to predict what is happening with DP2. Either revise the relevant sentences to not discuss the DP2 release date or revise the DP2 release date to be a realistic date. The latter would be preferable thinking about community planning.
- Line 195- I think when LSSTCam is first mentioned there should be references to the Gilmore et al. (2016) and Gilmore et al. (2018) SPIE publications.
- Line 246 - I thought DP0 is comprised of DP0.2 and DP0.3 and that DP0.2 is the Dark Energy Science Collaborations DC2 release. In that case I would expect there is a reference to the DESC DC2 paper (DESC 2021) https://ui.adsabs.harvard.edu/abs/2021ApJS..253...31L/abstract A suggestion, it might be helpful to move to the end of this section the link/reference to online documentation so a reader skimming the paper can easily find the documentation link.
- Figure 2 is referenced in the text before Figure 1 is.
- Line 535 - "Future campaigns will benefit from more stable and uniform flat fielding using the Rubin flat field system, described in P. Fagrelius & E. S. Rykoff (2025)" - There are no future campaigns with ComCam on the Simonyi Telescope.
- Wide Fast Deep - Bianco et al (2022) writes Wide-Fast-Deep and Ivezić et al. (2019) writes "Wide Fast Deep". Rubin Observatory papers are inconsistent so I raise it for the authors to decide.

Comment by Leanne Guy [24/Oct/25]

Additional “minor comments” in other sections:

- Line 1782 AU--> au
- Line 2535 - A one paragraph sentence. Perhaps this can be added to other paragraphs in the section.

Comment by James Mullaney [16/Dec/25]

My record of responses I’ve ID’d Sections 1 & 2:
>> Line 266 - Currently DP1 is under a two-year proprietary embargo period. How is the DP1 data going to be made accessible to the public? What is

the delivery method?

Done. Added:
”All Rubin Data Releases and Previews are subject to a two-year proprietary period, with immediate access granted exclusively to LSST data rights holders \citep[][]
{rdo-013}.
Data rights holders\footnote{See \url{https://www.lsst.org/scientists/international-drh-list}} are individuals or institutions with formal authorization to access proprietary data collected by the Vera C. Rubin Observatory.
After the two-year proprietary period, DP1 will be made public,
However, even once the data become public, access for non–data rights holders will not be provided through Rubin Data Access Centers in the US and Chile \citep[][]{rdo-013}

.
Alternative access mechanisms are still under discussion and have not yet been finalized.”

Section 2:

>> Some more detail about the observing campaign, how locations on sky were selected, and how ComCam observations were deemed "most useful for the community" to be included in DP2 is needed in this section. I provide some specifics questions and comments below.

Done. Added:
To support these goals, seven target fields were selected that span a range of stellar densities, overlap with external reference datasets, and collectively span the full breadth of the four primary \gls
{LSST}

science themes.

> > Add a summary of how the observations were taken, the planned cadence of observations, filters available. Some understanding of the time separation between observations on-sky in the same night, and how often observations were repeated would be useful in addition to the contents of Table 3 and Figures 4 and 5. Is AOS commissioning mostly taking images at the same spot across different airmasses?

Is this addressed? Personally, I don’t feel that this can be easily summarized in a way that would be useful. There is too much variation, and what do they mean by “how often observations were repeated?” How many failed and started again (why is this useful?) or how many times was a field observed with a particular filter (data already given)? And I feel we should steer away from talking more about AOS in this science release paper.

>> Please include the actual dates of the on-sky observing campaigns (start and end of LSSTComCam on-sky observations) early on in this Section rather than the reader having to convert the MJD dates in Figure 5 to get to UT calendar dates.
Done. This is now included in Section 1:
”DP1 is based on the reprocessing of 1792 science-grade exposures acquired during the first on-sky commissioning campaign, conducted over 48 nights between 2024-10-24 and 2024-12-11.”

>> Why were the specific pointings selected to make DP1? To test the Solar System Processing pipeline (I presume as a low ecliptic field), Seagull and 47 Tucanae to test dense sky backgrounds? Some clarification would be helpful to those who will be using this dataset in the future. How do these fields collectively span the full breadth of the four primary LSST science themes?

Done. Added:
To support these goals, seven target fields were selected that span a range of stellar densities, overlap with external reference datasets, and collectively span the full breadth of the four primary LSST science themes.

>> Figure 5 - I am finding Figure 5 difficult to parse when comparing to Table 1. From my count it looks like Rubin SV 38 7 has only 10 observations for the number of markers I can count, but Table 1 shows that 159 observations were taken. It also looks like SV 95 -25 gets most of the observations from Figure 5 but from Table 1 it is actually ECDFS that gets the most observations.

I agree with this setiment, but I don't think we should amend the figure. The figure has too low a resolution to be able to see individual points, the best it can do is indicate the days on which a given field was observed with a given filter. I feel this should be explained in the text and the figure caption.

>> Figure 3 - The transmission curves for ComCam are only provided in this paper. The values should be also provided in a supplemental data files or as data behind the figure.

The transmission curves are provided with DP1. We state this in Section 3, but we should also add a comment saying such in Figure 3's caption.

>> Line 549 - "..accumulating over 200 visits per to evaluate deep coadded images with integrated exposures roughly equivalent to those of the planned LSST Wide Fast Deep 10-year depth" - What are those expected depths? It would be helpful to remind the reader so they can compare to the limiting magnitude of the coadded observations in Table 2.

We have added the 10-year depth to Table 2.

>> Table 2 - it would be helpful to add in what the total on-sky exposure time is for the co-added observations per filter per field.

Since all the exposures that went into each coadd are 30s long, then it is straightforward to calculate this from table 1. As such, we don't feel we need to add this number.

>> A median 5-sigma per field per filter single PVI (processed visit image) source detection limits/limiting magnitude should be provided in a table.

This is now shown in Fig. 8 in Section 2.

>> It would be helpful to provide in some table per field per filter what the saturation apparent magnitude is.

The saturation apparent magnitude is independent of field. Presumably they want this for a point source? Do we include this information?

>> The galactic and ecliptic latitudes for each field should be listed in Table 1 or a separate table.

Why? This can be easily calculated from the RA and Dec by someone who is interested in this information.

>> What were the center pointings for the 4 Rubin_SV_38_7 field pointings? Was the center reported in Table 1 then offset by ~0.65 degrees with small random translational dithers?

I think this still needs to be included. I can't find it in Section 2.4.

>> It would be useful to discuss the airmass distribution of the visits in the text. Maybe the min/max airmass per filter per pointing could listed in a table or at

I think this needs adding.

>> Figure 6 is not referenced in the text. This should be presented/discussed in Section 2.4

Done. The following has been included at the end of Section 2.4:
Figure 6 shows sky coverage maps showing the distribution of visits in each of the seven DP1 fields, color coded by band. The images clearly show the focal plane chip gaps and dithering pattern. Only the detectors for which single frame processing succeeded are included in the plots, which explains why the central region of 47_Tuc looks thinner than the other fields.

>> Section 2.5 - Only median values and best value for seeing were reported in this section. What's the distribution of delivering seeing quality per filter per field? I see some of this information per filter is actually available in Figure 7, but Figure 7 is not referenced in the text. It would be helpful to get a sense of what the required delivering seeing for Rubin Observatory science observations to give a sense of the quality of the data/science return for DP1 compared to the LSST.

Figure 7 is now referenced in the text, with the requested information included:
Figure 7 shows the distribution of PSF FWHM (arcsec) over all 16071 individual sensors images. Ongoing efforts aim to quantify all sources of image degradation, including contributions from the camera system; static and dynamic optical components; telescope mount motion; observatory-induced seeing from the dome and primary mirror; and atmospheric conditions. For the LSST, the design specification for the median delivered image quality, referenced to the zenith and 550 nm, is 0.7.
This value corresponds to the measured median atmospheric seeing at the Cerro Pachon site (what is this number? 0.6?) and a system contribution to delivered image quality of 0.35 added in quadrature.

>> I presume fraction of sensors means over CCDs rather than the whole focal plane. Please clarify in the caption for Figure 7.

Done. The caption now reads:
Histograms showing the distribution of delivered image quality for all \nvisitdetectorsummaries single-epoch individual sensors in the DP1 dataset per passband (a) and per field (b). The median values are given in the legend.
>> The cumulative seeing plot per filter is useful, but please also add a histogram of the seeing distribution as a non-cumulative distribution as well.

Including both seems overkill.

Comment by James Mullaney [16/Dec/25]

>> Several times in the paper Data Preview (DP2) is referenced as planed to be released in mid-2026. My understanding from reviewing the Rubin Community Workshop presentations (https://rubinobservatory.org/events/rcw-2025) and the Schwamb et al (2025; https://ui.adsabs.harvard.edu/abs/2025RNAAS...9..244S/abstract) is that this is potentially no longer the case given the lessons learned for DP1 processing. The community will try to read tea leaves from the paper text to predict what is happening with DP2. Either revise the relevant sentences to not discuss the DP2 release date or revise the DP2 release date to be a realistic date. The latter would be preferable thinking about community planning.

We have removed reference to DP2 release dates.

>> Line 195- I think when LSSTCam is first mentioned there should be references to the Gilmore et al. (2016) and Gilmore et al. (2018) SPIE publications.

?

>> Line 246 - I thought DP0 is comprised of DP0.2 and DP0.3 and that DP0.2 is the Dark Energy Science Collaborations DC2 release. In that case I would expect there is a reference to the DESC DC2 paper (DESC 2021) https://ui.adsabs.harvard.edu/abs/2021ApJS..253...31L/abstract

Reference added.

>> A suggestion, it might be helpful to move to the end of this section the link/reference to online documentation so a reader skimming the paper can easily find the documentation link.

Has this been addressed?

>> Figure 2 is referenced in the text before Figure 1 is.

This have been corrected.

>> Line 535 - "Future campaigns will benefit from more stable and uniform flat fielding using the Rubin flat field system, described in P. Fagrelius & E. S. Rykoff (2025)" - There are no future campaigns with ComCam on the Simonyi Telescope.

Just add “with LSSTCam”?

>> Wide Fast Deep - Bianco et al (2022) writes Wide-Fast-Deep and Ivezić et al. (2019) writes "Wide Fast Deep". Rubin Observatory papers are inconsistent so I raise it for the authors to decide.

?

Comment by James Mullaney [16/Dec/25]

>> Line 1782 AU--> au

Done

>> Line 2535 - A one paragraph sentence. Perhaps this can be added to other paragraphs in the section.

A paragraph is defined as a piece of text that deals with a single point or theme. The number of sentences it contains is inconsequential. (Let's not write that! 🙂)

Comment by Leanne Guy [14/Jan/26]

James Mullaney Agree – but I just joined the single sentence to the preceding paragraph

Comment by Leanne Guy [14/Jan/26]

Line 195: Camera team say that this reference is outdated so I will not include it.

Line 246: Link to DP0 is included as a footer

All others done

Comment by Leanne Guy [14/Jan/26]

Done

Comment by Leanne Guy [14/Jan/26]

James Mullaney
>> Line 195 : Aaron confirms that this reference is out of date and recommended another. I will defer to the Camera lead's judgement on this.

>> A suggestion, it might be helpful to move to the end of this section the link/reference to online documentation so a reader skimming the paper can easily find the documentation link.

Has this been addressed? Yes

Just add "with LSSTCam"? Yes

Wide fast Deep – Wide-Fast-Deep. Glossary was updated and this now comes up automatically in all papers

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